# 7. Deep Learning 1

Machine learning

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# Introduction

- Differences between shallow and deep ANN
- Differences between ANN and CNN
  - concept
- Basic building blocks
  - Convolution layer
  - Activation layer (ReLU, Softmax)
  - Pooling layer
  - Fully Connected layer
  - DropOut layer
  - BatchNorm layer
- Basic CNN architecture
- Deep learning frameworks

# Differences between shallow and deep ANN

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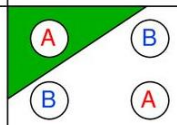
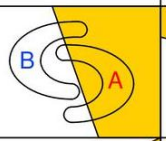
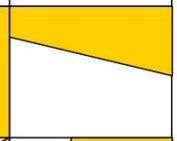
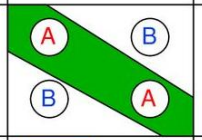
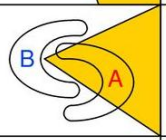
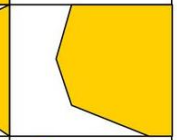
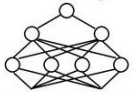
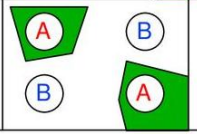
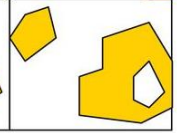
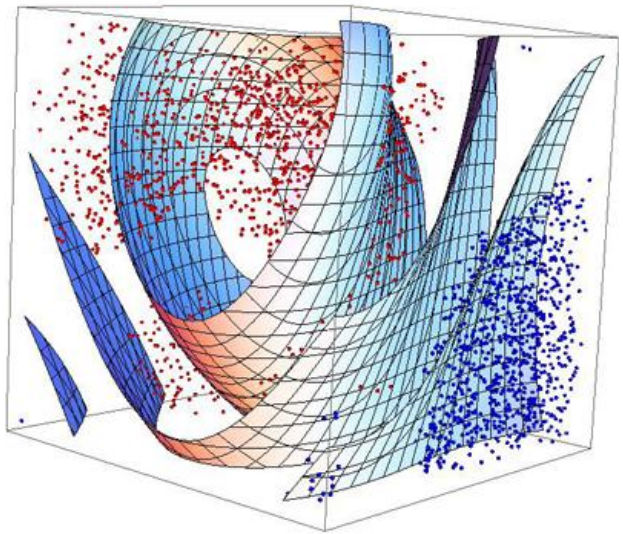
# Differences between shallow and deep ANN

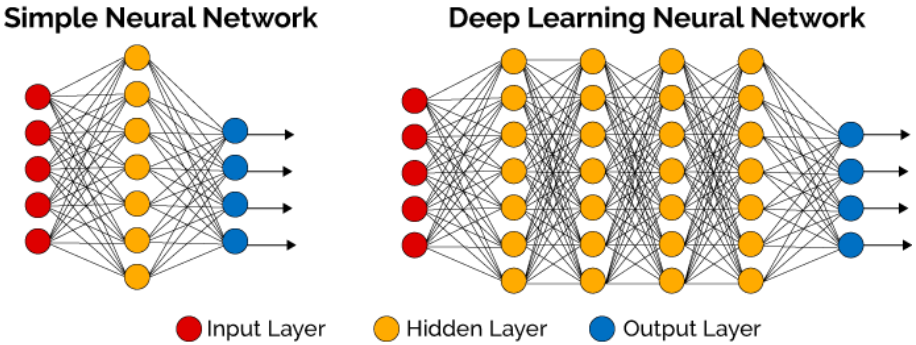
- earlier only shallow networks with max. 3 layers were considered (theoretically every N-dimensional convex function should be possible to approximate by 2-layer network (non-convex 3-layer) with enough neurons – division of N-D space by thus hyperplane

| Structure | Types of Decision Regions | Exclusive-OR Problem | Classes with Meshed regions | Most General Region Shapes |
|---|---|---|---|---|
| Single-Layer  | Half Plane Bounded By Hyperplane |   |   |  |
| Two-Layer  | Convex Open Or Closed Regions |   |   |  |
| Three-Layer  | Arbitrary (Complexity Limited by No. of Nodes) |   |   |  |



# Differences between shallow and deep ANN

- Nowadays, it is turning out that the deeper architectures brings higher level of abstraction by combining low-level features (formed in first layers) into the high-level features – it brings higher performance than shallow networks.
- Drawback:
  - huge increase of parameters (higher data, memory and computational requirements and overfitting problem)
  - vanishing of activation during forward pass and gradient during backpropagation – can be solved by new techniques (BatchNorm, weight initialization, ReLU activation function)



# Differences between ANN and CNN

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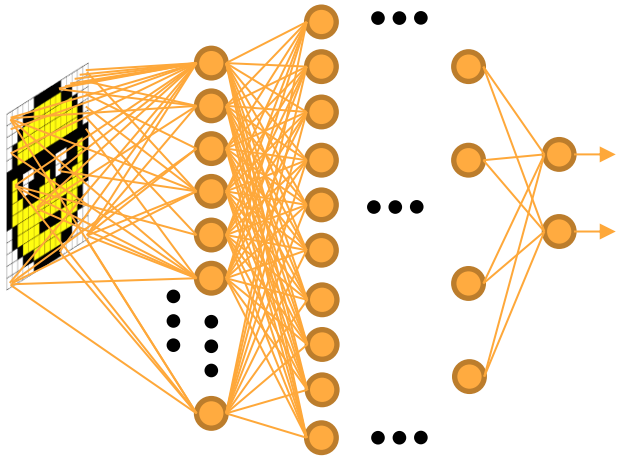
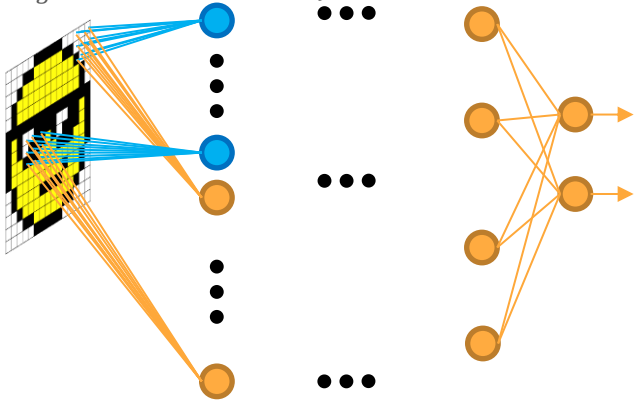
# Differences between ANN and CNN – concept

- ANN – fully connected
  - each neuron is connected to each neuron from previous layer
  - each weight of each neuron is independent
  - $N_{weights}^{neuron} = N_{inputs}$ $(200 \times 200 \times 3 = 120{,}000)$
  - $N_{weights}^{layer} = N_{inputs} \cdot N_{neurons}$ $(120{,}000 \cdot 50 = 6M)$



- CNN
  - each neuron is connected to the local neurons from previous layer – space dependency?
  - each neuron shares weights with others – want the same local feature over all input
  - $N_{weights}^{neuron} = N_{local\_conn}$ $(3 \times 3 \times 3 = 27)$
  - $N_{weights}^{layer} = N_{local\_conn} \cdot N_{features}$ $(27 \times 2 = 54)$



# Basic building blocks

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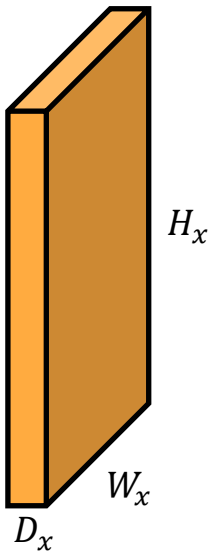
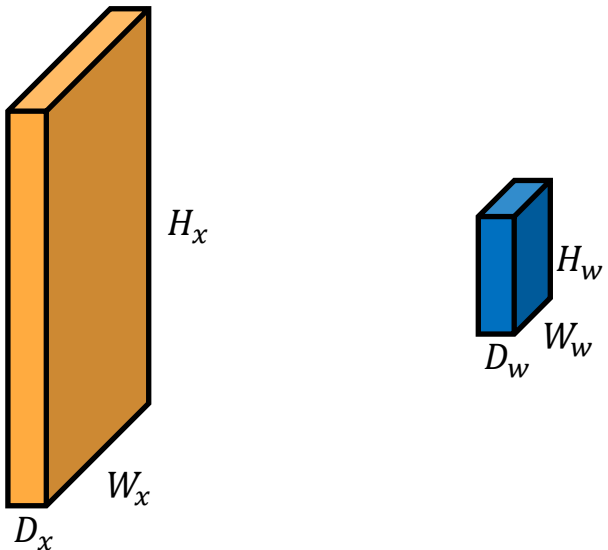
# Basic building blocks – Convolutional layer

- input image $\boldsymbol{x}$ with size $W_x \times H_x \times D_x$ – Width, Height, Depth
  - size of input image, depth is number of channels (grayscale, RGB, multimodal image, etc.)



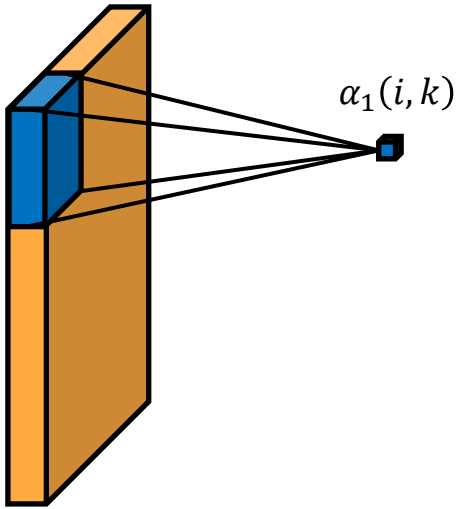
# Basic building blocks – Convolutional layer

- filter ***w*** with $W_w \times H_w \times D_w$ – Width, Height, Depth – usually square shape $W_w = H_w = F$
  - size of filter – local connection, depth – always the same as depth of image $D_w = D_x$



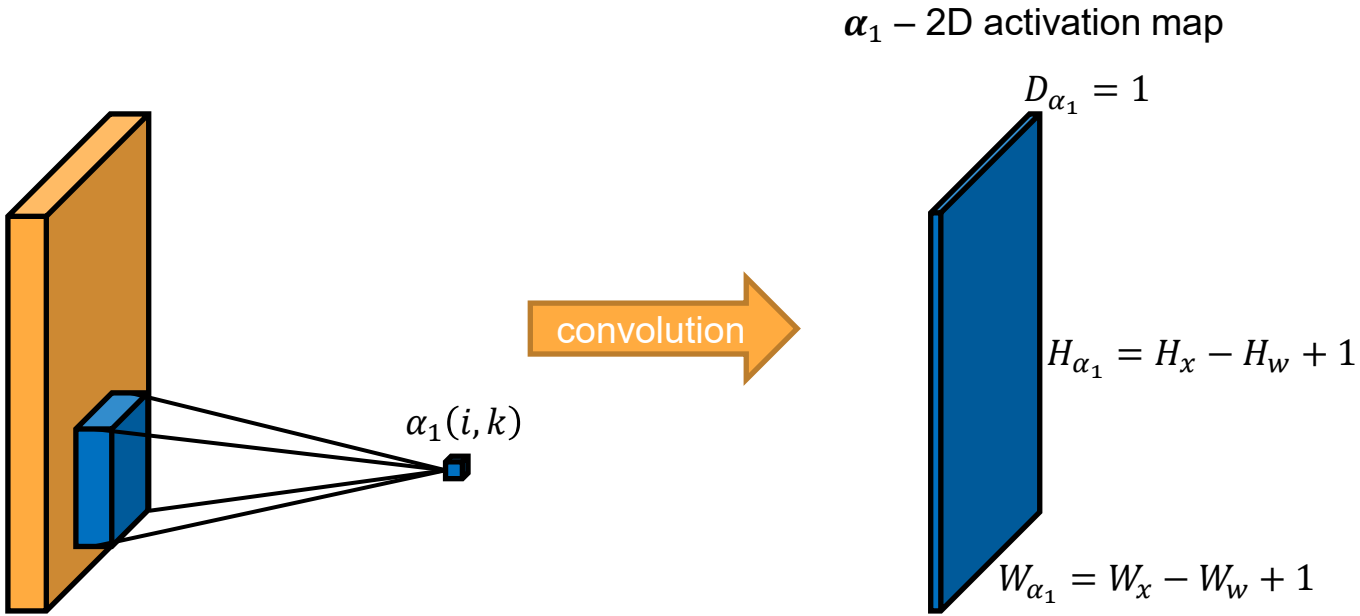
# Basic building blocks – Convolutional layer

- compute dot product between part of the input image and filter
  - **convolution** for one image position: $\alpha(i,k) = \sum_{m=-W_w/2}^{W_w/2} \sum_{n=-H_w/2}^{H_w/2} x(i-m, k-n) w(m,n)$
    … something familiar?
  - multiplication and final sum – **same as neuron activation**: $\sum_{ind=1}^{W_w H_w} x_{ind} w_{ind}$



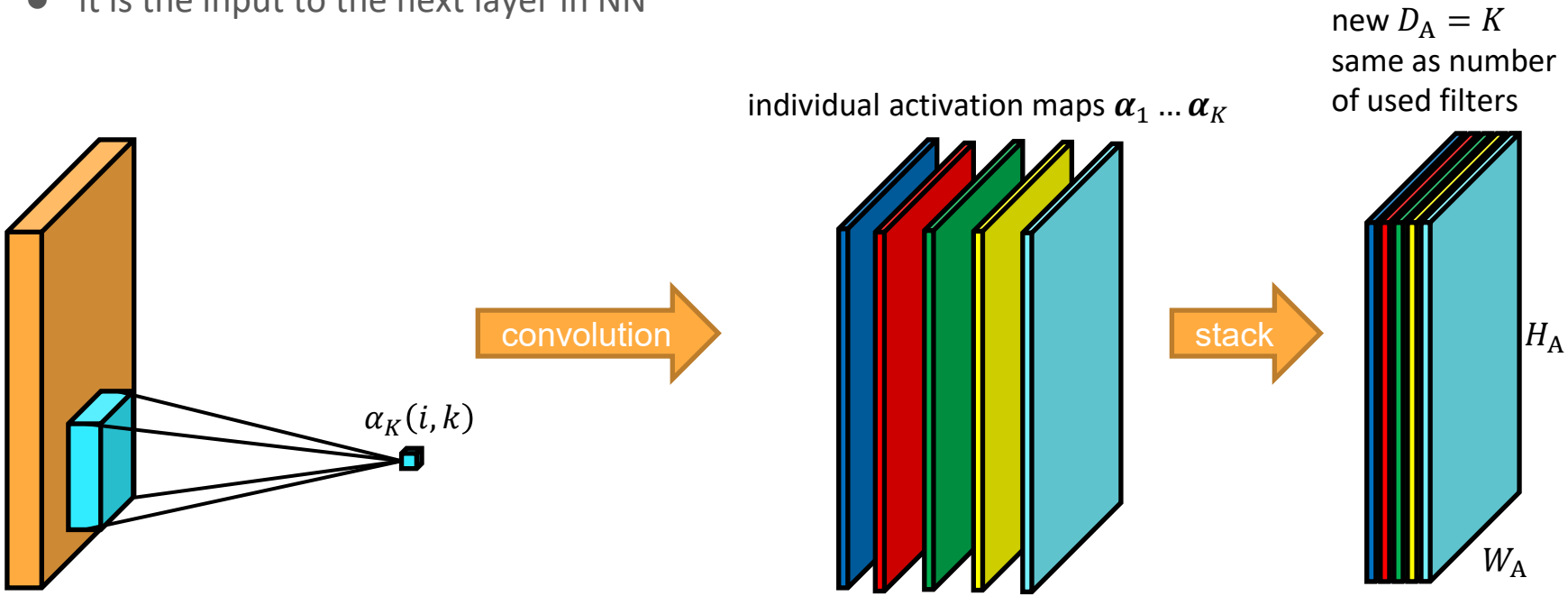
# Basic building blocks – Convolutional layer

- compute convolution over all image – activation map of smaller size is obtained



# Basic building blocks – Convolutional layer

- compute convolution over all image with another $K$ filters – new activation maps $\boldsymbol{\alpha}_1 \ldots \boldsymbol{\alpha}_K$
- output of each convolutional layer is new 3D activation map $\mathbf{A}_l$
- it is the input to the next layer in NN

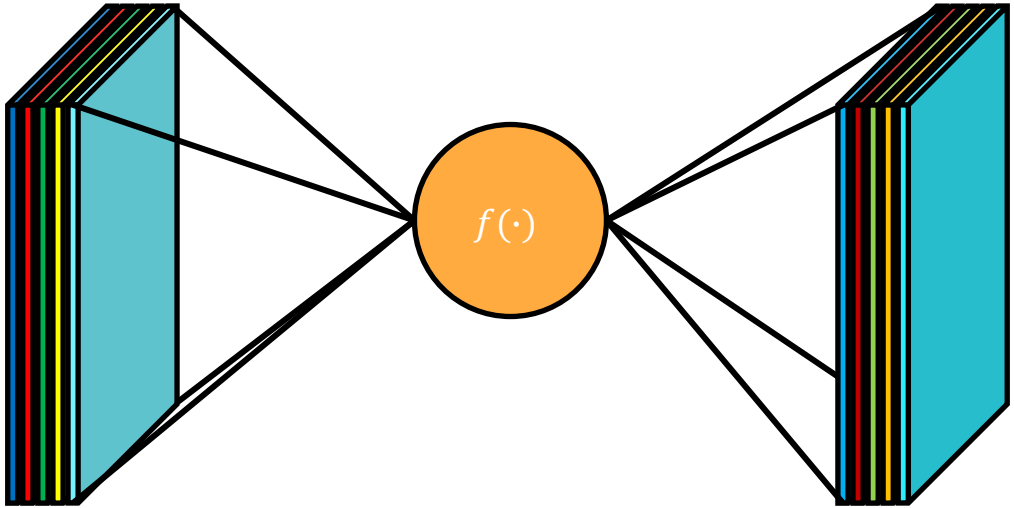


# Basic building blocks – Convolutional layer

- Hyperparameters of convolutional layer:
  - number of filters in layer $K$
  - their spatial extent $F$ (usually odd number)
  - the stride $S$ – step of “convolution”
  - the amount of zero padding $P$ – extension by zeros at border
- Input size $W_l \times H_l \times D_l$ –> output size $W_{l+1} \times H_{l+1} \times D_{l+1}$
  - $W_{l+1} = \frac{W_l - F_l + 2P_l}{S_l} + 1$
  - $H_{l+1} = \frac{H_l - F_l + 2P_l}{S_l} + 1$
  - $D_{l+1} = K_l$
- All filters in the same layer must have the same size
- Output size must have integer size in each dimension
- Number of parameters per layer is $F_l F_l D_l K_l$ weights $+K_l$ biases

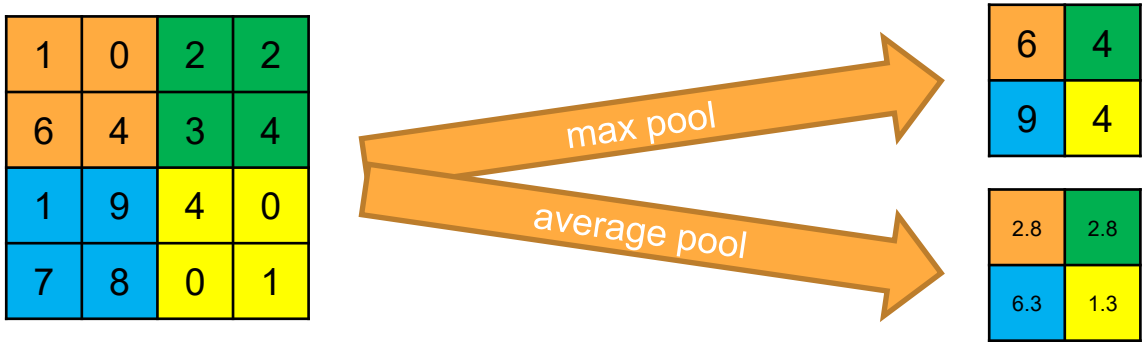
# Basic building blocks – Activation layer

- Input: 3D activation map $\mathbf{A}_l$
- Output: 3D feature map of the same size as $\mathbf{A}_l$.
- Same as in ANN – point-by-point transformation of activation-by-activation function $f(\mathbf{A}_l)$
- Usually ReLU, or Softmax as output layer – without any hyperparameter



# Basic building blocks – Pooling layer

- Input: image/activation map/feature map
- Output: shrink input in Weight and Height direction, Depth remains unchanged
- Max pool (more often) or average pool
- Transferring the strongest (max pool) activation on output
- Lower the parameters needed in the deeper layers

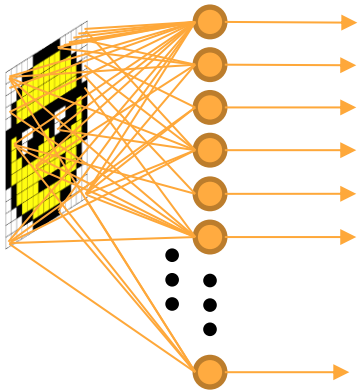


# Basic building blocks – Pooling layer

- Hyperparameters of pooling layer:
  - their spatial extent $F$ (usually $2$ or $3$)
  - the stride $S$ – step moving window (usually $2$)
- Input size $W_l \times H_l \times D_l$ –> output size $W_{l+1} \times H_{l+1} \times D_{l+1}$
  - $W_{l+1} = \frac{W_l - F_l}{S_l} + 1$
  - $H_{l+1} = \frac{H_l - F_l}{S_l} + 1$
  - $D_{l+1} = D_l$
- Output size must have integer size in each dimension
- Number of learned parameters per layer is zero

# Basic building blocks – Fully connected layer

- Input: image/activation map/feature map
- Output: vector or scalar value
- The same as standard ANN layer
- Activation function ReLU/Softmax (output layer)



- Hyperparameters of fully connected (FC) layer:
  - number of neurons in layer $K$
- Input size (vectorized) $W_l H_l D_l$ –> output size $K_l$
- If two subsequent FC layers – $K_{l-1}$ –> output size $K_l$
- Number of parameters in layer $W_l H_l D_l K_l + K_l$

# Basic building blocks – DropOut layer

- The same function as in standard ANNs
- Disable some outputs from neurons in convolutional or fully connected layers
- Realized as independent layer that is used in the training phase
- Put before conv. or FC layer

- Hyperparameters of DropOut layer:
  - probability that neuron will not be disabled $p$
- Input size $W_l \times H_l \times D_l$ –> output size $W_l \times H_l \times D_l$
- No learned parameter

# Basic building blocks – BatchNorm layer

- The same function as in standard ANNs
- Normalize inputs to the next convolutional or fully connected layer
- Realized as independent layer – works independently for each filter (feature dimension)
- Put before conv. or FC layer

- Hyperparameters of BatchNorm layer:
  - exponential decay $\alpha$
- Input size $W_l \times H_l \times D_l$ –> output size $W_l \times H_l \times D_l$

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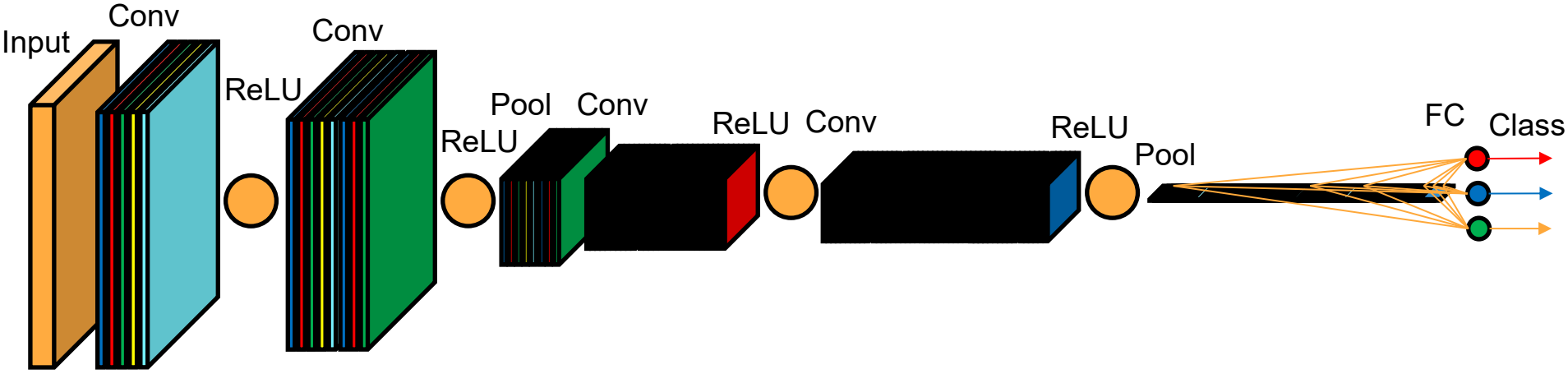
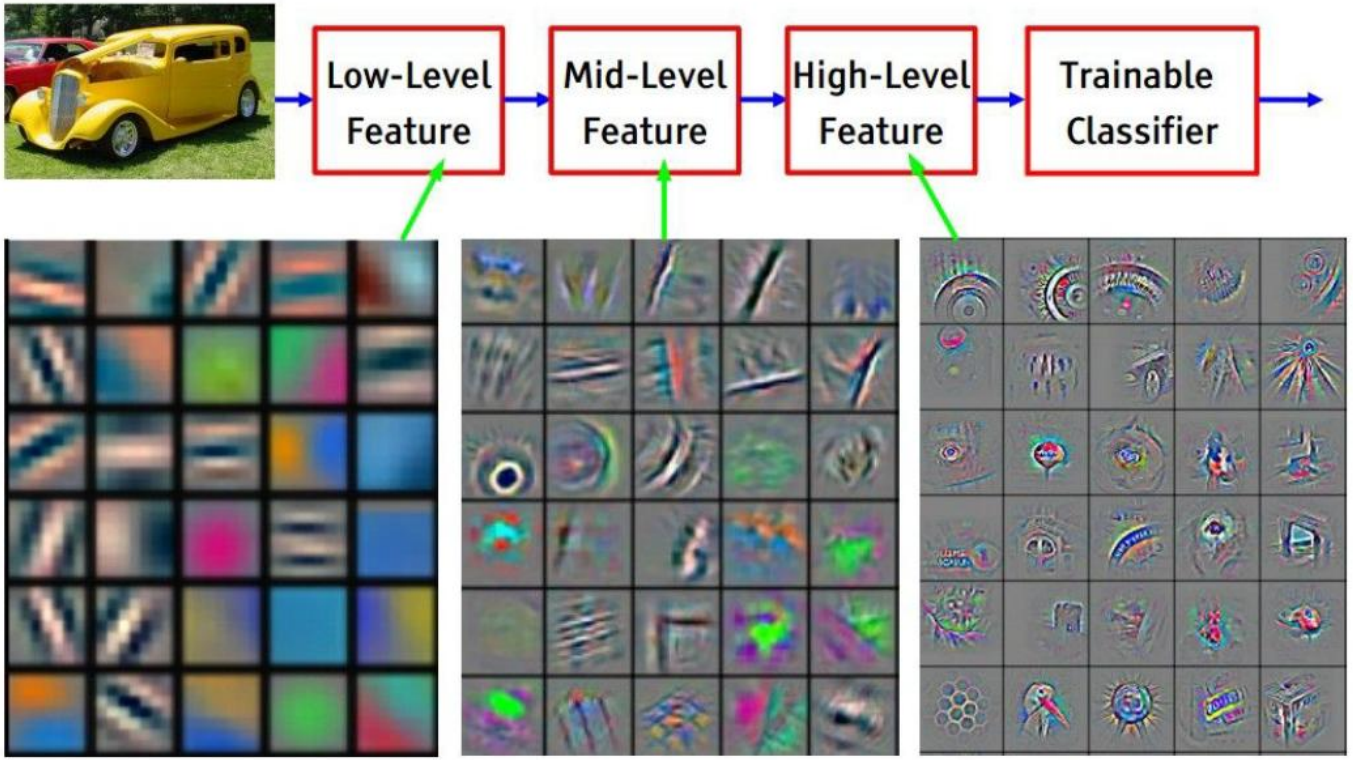
- Usually a few combination of convolution and ReLU activation layers is followed by max pooling layer – this is repeated a few times – at the end, 1-3 fully connected layers are used, when the first with ReLU activation functions and the last output FC layer with Softmax activation function.
- Generally, as the input image goes through CNN, its size is gradually reduced in the Width and Height directions, while the Depth is growing – at the end 1D feature vector is classified by output FC layer.



# Basic CNN architecture – feature visualization




Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

# Basic CNN architecture – filters and activation map